# HALEEMA MALIK

## About me

A self-motivated, passionate, and enthusiastic individual aspiring to pursue a career in Intelligent Interactive Systems. I am a recent graduate of BS (Hons) Software Engineering from Liverpool John Moores University (LJMU), with First-Class Honours. I have a strong ambition to enhance accessibility and personalized learning, with a particular focus on supporting children with disabilities in decision-making applications, while bridging the gap between technology, education, and inclusivity.


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## EDUCATION

### BSc. Hons in Software Engineering

Liverpool John Moore's University (LJMU)

*09/2022 – 07/2025*

*Main Courses:*

- **Introduction to Programming, Data Structures and Algorithms, Computer Systems, Database Systems, Object Oriented Systems, Software Engineering Principles, Automata, Languages and Computation, Data Modelling**

## HONOR AWARDS

Gold Medalist Distinction Holder (2022 - 2023)

*Liverpool John Moore's University*

- Was able to achieve straight distinctions for (First year) academic year 2022-23 in a class of 65 students at LJMU (Liverpool John Moore's University), which was ranked 3rd for Course Quality in the University Compare 2023 Rankings.

Gold Medalist Distinction Holder (2023 - 2024)

*Liverpool John Moore's University*

- Achieved straight distinctions for the academic year 2023-24 (Second year) in a class of 63 students at LJMU (Liverpool John Moore's University), which was ranked 2nd for Course Quality in the University Compare 2024 Rankings.

## SCHOLARSHIPS

Nanodegree AWS AI and ML Scholarship Program (09/2024 - Present)

*It is a competitive learning scholarship program for AI and ML enthusiasts, sponsored by Amazon Web Services (AWS) to get hands-on experience from experts in the field for industry-recognized courses and projects.*

Bachelor's Scholarship (09/2022 – 07/2025)

*Awarded a 40% scholarship on college tuition fees in 2022, among 1,000+ accepted students, in recognition of outstanding academic performance and exceptional high school grades.*

## RESEARCH EXPERIENCE

### Research Associate

All Pakistan Student Union (APSU)

*09/2024 – 05/2025*

*Tasks*

- Worked with **Dr. Sidra** in building human emotion recognition with CNNs and deep learning models to enhance the accuracy of emotion classification for patients' conditions.
- Optimizing multimodal data, including facial expressions, and voice tones for real-time emotion analysis.

### Research Intern

THE CENTRE OF EXCELLENCE FOR TECHNOLOGY QUANTUM AND ARTIFICIAL INTELLIGENCE PAKISTAN (CETQAP)

*11/2024 – 04/2025*

*Tasks*

- Working under the supervision of **Dr. Zuhair Ahmed** to explore and understand the principles of Quantum Engineering.
- Contributing to **Pakistan's first and only Quantum Institute**, playing a pioneering role in advancing quantum science in the region.
- Engaged in technical experiments and simulations to analyze quantum behaviors in engineered systems.
- Participating in seminars and technical discussions to stay updated with advancements in global quantum research.

## RESEARCH EXPERIENCE

### Contributor/ Learner

Beaconhouse International College (BIC) Core Research Program

*12/2023 - 02/2024*

*Achievements/Tasks*

- Acquired in-depth knowledge on how to craft and write comprehensive research papers. This included selecting pertinent topics, conducting literature reviews, formulating research questions, and applying appropriate methodologies.
- Presented my research findings to the CEO and other senior staff members at Beaconhouse international college (BIC). This experience honed my ability to communicate complex research concepts clearly and confidently to a professional audience, enhancing my public speaking and presentation skills.

## PROFESSIONAL EXPERIENCE

BACKEND INTERN- CAPREGSOFT (PRIVATE) LIMITED (04/2025 - 05/2025)

*I gained hands-on experience in developing server-side applications using GO (Golang). My role involved building and optimizing APIs, ensuring high performance and scalability. I also collaborated with development team to integrate backend with frontend applications.*

INTERN TRAINEE - DIGIWOP (PRIVATE) LIMITED (08/2024 - 09/2024)

*Assisted in designing and maintaining websites, ensuring responsive layouts and optimized user experiences. Collaborated with the marketing team to develop and implement digital marketing strategies, including content creation and social media management.*

HOME TUTOR (10/2024 - 03/2025)

*Worked as a Home Tutor for Mathematics and Computer Science (Grades 10–12), providing personalized lessons, exam preparation, and concept reinforcement. Assisted students in understanding core principles, improving problem-solving skills, and achieving higher academic performance through tailored teaching methods.*

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DEVELOPER -GREEN X SOLUTIONS (03/2024 - 05/2024)

*Engaging clients to understand their requirements for website design and functionality. Designing and developing the front-end of websites. Evaluating website for better performance.*

## PROJECTS

### NFT Music Price Prediction

- I developed an NFT Music Price Predictor in Python that uses machine learning to estimate the last sale price of music NFTs.
- The project applies ensemble models like Random Forest and AdaBoost, evaluated with metrics such as $R^2$, RMSE, and F1 score.
- It is done in four phases—starting with NFT metadata, then adding collection data, owner data, and finally combining all features—the models (AdaBoost and Random Forest) demonstrated that AdaBoost excelled in early phases, while Random Forest delivered superior regression performance in Phases 3 and 4, and classification in Phase 4 also achieved strong ROC AUC and F1 scores

### Predicting Bike Sharing Hype Demand

- Implemented bike rental demand forecasting using AutoGluon's TabularPredictor, training an initial baseline model on the Kaggle dataset.
- Enhanced performance through feature engineering, extracting time-series attributes (hour, day, month) from datetime fields.
- Improved model accuracy via hyperparameter tuning across multiple runs and evaluated incremental gains at each stage (baseline → engineered features → tuned model).

### Dog Breed Identification Using CNNs

- Using CNN models to correctly identify images of dogs and their breeds, trained on the ImageNet dataset, with three different architectures: AlexNet, VGG, and ResNet.
- The program uses the time module to calculate the runtime and compare the accuracy of the results across the three architectures.
- Leveraged cloud-based solutions (AWS S3) for storing and processing dog images securely and classifying them.

### Smart Irrigation System

- Designed and implemented an Arduino-based automated irrigation system that monitors soil moisture via analog sensors and triggers a relay module to control a DC water pump when moisture drops below a defined threshold.
- Integrated real-time feedback and logging: moisture levels are classified (Dry, Moist, Wet), displayed on a 16×2 LCD and LEDs, and logged to an SD card for historical data analysis.
- Programmed in C++, managed circuit design (soil sensor, relay, power supply), and ensured embedded control loop timing for reading sensor data every 5 seconds for responsive actuation.

## LEADERSHIP & PROFESSIONAL ACTIVITIES

President of Computer Science Society
(09/2023 - 05/2024)
*Student council*

- Successfully organized hackathons, Google Developer Student Club (GDSC) workshops, and industrial visits, ensuring seamless execution and participant engagement.

Vice President for SDG's Research Unit
(07/2024 - Present)
*Student council*

- Organized plantation campaigns with 100+ participants to plant new seeds, promoting sustainable environmental goals.
- Collaborated with the All-Pakistan Student Union (APSU) to educate villagers on sustainability and best practices for environmental conservation.

Campus Ambassador (09/2024 - 12/2024)
*Girl Script Summer of Code*

- Encouraged students to explore new resources and open-source technologies, guiding them on how to contribute effectively to future learning and career opportunities.

Badminton Team Lead (07/2024 - 09/2024)
*BIC -Beaconhouse International College*

- Secured victory in the IVY League Badminton Tournament 2024 as the best player and team leader, demonstrating exceptional skill and leadership to achieve success.

## CERTIFICATIONS

Programming with Python - Harvard University

- Learned core programming concepts (OOP, data structures, file handling) and applied Python for problem-solving, automation, and data analysis with hands-on projects.

Software Practices Bootcamp - CapregSoft

- Gained practical experience in software engineering workflows, Git/GitHub, Agile methods, debugging, and unit testing, improving collaborative, parallel programming and coding practices.

AI Programming with Python - Udacity

- Built a foundation in machine learning pipelines, NumPy, Pandas, Matplotlib, and scikit-learn, applying ML algorithms to real-world datasets.

Tokenization to Vector Quantization - DeepLearning.AI

- Explored NLP tokenization, embeddings, and vector quantization techniques, enhancing understanding of text representation for large language models.

LangChain for LLM Application Development - DeepLearning.AI

- Learned to build LLM-powered applications using LangChain, focusing on prompt chaining, retrieval-augmented generation (RAG), and API integration.

## VOLUNTEERING AND PROFESSIONAL MEMBERSHIP

**VOLUNTEER STUFF - POF HOSPITAL**
**MEMBER OF WOMEN TECHMAKERS - GOOGLE DEVELOPERS PROGRAM**
**MEMBER - GOOGLE DEVELOPERS CLUB**

## LANGUAGES

English
*Full Professional Proficiency*

Urdu
*Native or Bilingual Proficiency*

Punjabi
*Native or Bilingual Proficiency*

## TECHNICAL SKILLS

**Programming Languages and Frameworks:** Python, Java, C++, JavaScript, HTML, CSS, React.js, Node.js, PHP, GOLANG, Flask, TensorFlow, Keras

**Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Power BI, MySQL, MongoDB, Firebase, Scikit-learn, AWS, Excel